THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE, REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

LONGIDO DISTRICT, FORM FOUR PRE – MOCK EXAMINATION SYNDICATE

CHEMISTRY – MARKING SCHEME

1. ***(01 Mark @ = 10 Marks)***

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
D	A	C	C	B	A	E	D	D	E

2. ***(01 Mark @ = 05 Marks)***

(i)	(ii)	(iii)	(iv)	(v)
G	H	A	C	B

3. (a) A – Graphite ***01 Mark***

B – Amorphous ***01 Mark***

C – Diamond ***01 Mark***

(b) (i) Allotrope B is formed by incomplete combustion of material containing carbon. ***02 Mark***

(ii) Required any two difference between allotropes A and C. ***@ = 0.5 Marks = 02 Marks***

Allotrope A (Graphite)	**Allotrope C (Diamond)**
It is soft and slippery	It is hard (the hardest natural substance known)
It has a low density	It has a high density
It is a good conductor of heat and electricity	It is a bad conductor of heat and electricity
It has a low melting point and boiling point	It has a high melting point and boiling point

4. (a) (i) Carbonates and hydroxides of potassium and sodium are strong on heat. They are not

Decompose on heat. ***01 Mark***

(ii) Because they have full filled electrons in their outermost shells. They have an octet structure. ***01 Mark***

(iii) Because oxygen supplied is not enough to complete burning of a fuel. ***01 Mark***

(b) (i) R.M.W of hydrated compound ($Na_2CO_3.5H_2O$) = 2(23) + 1(12) + 3(16) + 5(18) = 196. ***0.5 Marks***

R.M.W of water ($5H_2O$) = 5(18) = 90 ***0.5 Marks***

% composition by mass = $\frac{90}{196}$ x 100% ***0.5 Marks***

% composition of water in a compound = 45.9% ***0.5 Marks***

(ii) Required any two disadvantages of hard water. ***@ = 01 Mark = 02 Marks***

- Temporary hardness of water causes lime scale in water boilers.
- Hard water needs more soap than soft water.
- Hard water leaves scummy deposits on clothing and utensils that are difficult to wash it.

5. (a) Radical: is a group of atoms which acts as a single atom in forming a compound.

OR: is a group of atoms with unpaired electrons ***01 Mark***

(b) Required the empirical formula and molecular formula

(i) Empirical formula ***@ row = 0.5 Marks = 03 Marks***

Element symbols	C	H	O
% composition	68.54	8.63	22.83
R.A.M	12	1	16
$\frac{\%\ composition}{R.A.M}$	5.71	8.63	1.43
Divide by smallest number	3.99	6.03	1
Whole number ratios	4	6	1

Empirical formula of a compound is C_4H_6O ***01 Mark***

(ii) Molecular formula

R.M.M = n (sum of R.A.M) ***0.5 Marks***

140 = n (4(12) + 6(1) + 1(16))

140 = n (48 + 6 + 16)

140 = 70n

n = 2 ***0.5 Mark***

Then: Molecular formula = n (Empirical formula) ***0.5 Mark***

= 2 (C_4H_6O)

Molecular formula = $C_8H_{12}O_2$ ***0.5 Mark***

6. (a) (i) Modern periodic law

States that "The properties of an element are periodic function of their atomic number" ***01 Mark***

(ii) Periodicity: is the periodic repetition of similar properties when elements are arranged in order of increasing atomic numbers. ***01 Mark***

(iii) Electronic configuration: is the arrangement/distribution of electrons around the nucleus of an atom in closed orbits(shells). ***01 Mark***

(b) (i) Winnowing ***01 Mark***

(ii) Solvent extraction ***01 Mark***

(iii) Fractional distillation ***01 Mark***

(iv) Sublimation ***01 Mark***

7. (a) (i) To complete the table

Table JJ ***01 Mark***

EXPERIMENT	PILOT	1	2	3
Final burette reading (cm^3)	16.30	16.00	32.00	48.30
Initial burette reading (cm^3)	0.00	0.00	16.00	32.00
Volume of acid used (cm^3)	**16.30**	**16.00**	**16.00**	**16.30**

Average volume of acid = $\frac{16.00+16.00+16.30}{3}$ = 16.10 cm^3 ***01 Mark***

(ii) Balanced chemical equation

$2HCl(aq) + X_2CO_3(aq \longrightarrow 2XCl(aq) + H_2O(l) + CO_2(g)$ ***01 Mark***

Mole ratios: $n_a = 2, \quad n_b = 1$

(iii) Molarity of a metal carbonate

$$\frac{M_a v_a}{n_a} = \frac{M_b v_b}{n_b}$$

$$M_b = \frac{M_a v_a n_b}{v_b n_a}$$

$$= \frac{0.25 \text{ x } 16.1 \text{ x } 1}{25 \text{ x } 2}$$

Molarity of metal carbonate = $0.08 mol/dm^3$ ***01 Mark***

(b) (i) Required the relative atomic mass of a metal carbonate

$$\text{Concentration} = \frac{\text{Mass}}{\text{Volume of solution}} = \frac{2.12g}{0.25dm^3} = 8.48 g/dm^3$$

$$\text{Molar mass} = \frac{\text{Concentration}}{\text{Molarity}} = \frac{8.48 g/dm^3}{0.08 mol/dm^3} = 106 g/mol$$

Relative molecular mass = 106g/mol ***01 Mark***

(ii) Required the atomic mass of X.

2(X) + 1(C) + 3(O) = 106

2X + 12 + 48 = 106

2X + 60 = 106

2X = 46

X = 23

Atomic mass of X = 23g ***01 Mark***

(iii) Required the electronic configuration of the metal ion

A metal is Sodium (Na)

A metal ion is Na^+

Electronic configuration of a metal ion (Na^+) is 2:8 ***01 Mark***

8. (a) Required to complete and balance.

(i) $MgO(S) + H_2O(l) \longrightarrow Mg(OH)_2(aq)$ ***01 Mark***

(ii) $2AgNO_3(aq) + MgCl_2(aq) \longrightarrow 2AgCl(s) + Mg(NO_3)_2(aq)$ ***01 Mark***

(iii) $2SO_2(g) + O_2(g) \longrightarrow 2SO_3(g)$ ***01 Mark***

(iv) $NH_3(g) + HCl(g) \longrightarrow NH_4Cl(s)$ ***01 Mark***

(b) Balanced equation for decomposition of potassium chlorate

$2KClO_3(s) \xrightarrow{Heat} 2KCl(s) + 3O_2(g)$ ***01 Mark***

245g of $2KClO_3$ → $67.2dm^3$ of $3O_2$ at STP

12.25g of $2KClO_3$ → Y? of $3O_2$ at STP? ***01 Mark***

$Y = \frac{12.25g \times 67.2dm^3}{245g} = 3.36dm^3$

Volume of oxygen gas = $3.36dm^3$ ***01 Mark***

9. (a) Required to define:

(i) Activation energy: is the minimum amount of energy that must be supplied to the - reactants in a chemical reaction for them to react and form products. ***01 Mark***

(ii) Isolated system: is the system in which both energy and reactants are not interchanging With the surrounding. ***01 Mark***

(iii) Chemical kinetics: is the study of the rates of chemical reaction and the factors that – affects the rate of chemical reaction. ***01 Mark***

(b) Le charteliers' principle: States that "If a change is applied to a system in equilibrium, the equilibrium will shift to the direction opposite to the change applied" ***01 Mark***

(c) (i) Exothermic reaction ***01 Mark***

(ii) A – Nitrogen and hydrogen ***0.5 Marks***

B – Ammonia ***0.5 Marks***

C – Activation energy ***0.5 Marks***

D – Enthalpy change/ Heat change ***0.5 Marks***

10. (a) Faraday's first law of electrolysis states that:,

"The mass of a substance deposited or liberated during electrolysis is directly proportional-

to the quantity of electricity passed through the electrolyte". ***01 Mark***

Faraday's second law of electrolysis states that,:

"When the same quantity of electricity is passed through solutions of different electrolytes,

the mass of the substances deposited or liberated at the electrodes are directly proportional

to their chemical equivalents". ***01 Mark***

(b) Discharge equation for chlorine:

$2Cl^-(aq) \longrightarrow Cl_2(g) + 2e^-(aq)$ ***01 Marks***

If 2F of electricity $\longrightarrow$ 22.4 litres of Cl_2 at STP

F? of electricity $\longrightarrow$ 1.4 litres of Cl_2 at STP

22.4l x F = 2 x 1.4 Divide by 22.4l both sides

<u>Number of faraday liberated during electrolysis = 0.125F</u> ***01 Mark***

(i) Discharge equation for oxygen:

$4OH^-(aq) \longrightarrow 2H_2O(l) + O_2(g) + 4e^-(aq)$ ***0.5 Mark***

4F = 22.4L of oxygen at STP

0.125F = ?

$= \frac{22.4L\ X\ 0.125F}{4F}$ = 0.7L

<u>Volume of oxygen produced = 0.7 litres</u> ***01 Mark***

(ii) Discharge equation for copper:

$Cu^+(aq) + 2e^-(aq) \longrightarrow Cu(s)$ ***0.5 Marks***

2F = 63.5g of copper

0.125F = ?

$= \frac{63.5g\ x\ 0.125F}{2F}$ = 3.97g

<u>Mass of copper produced = 3.97g</u> ***01 Mark***

11. (a) (i) Fire fighting: is the process of extinguishing/putting off harmful fire. ***01 Mark***

(ii) Striking back of a flame: is the phenomena occur when a flame of a Bunsen burner-changes its direction and burn back to the jet below the barrel/chimney. ***01 Mark***

(b) Required to explain five effects of air pollution. ***@ = 01 Mark = 05 Marks***

(i) Health problems: Polluted air may affect the respiratory system, leading to cancer or death.

(ii) Global warming: This is the increase of the average temperature of the earth due to excessive emission of greenhouse gases such as carbon dioxide, sulphur dioxide and methane. These gases are air pollutants.

(iii) Depletion of ozone layer: Gases such as nitrogen monoxide, nitrogen dioxide, and Chlorofluorocarbon contribute to depletion of ozone layer which protects us from harmful ultraviolet rays from the sun.

(iv) Acid rain: Air pollution causes acid rain which destroys vegetation, kill animal and Plants and destroy art and architecture made by calcium and iron.

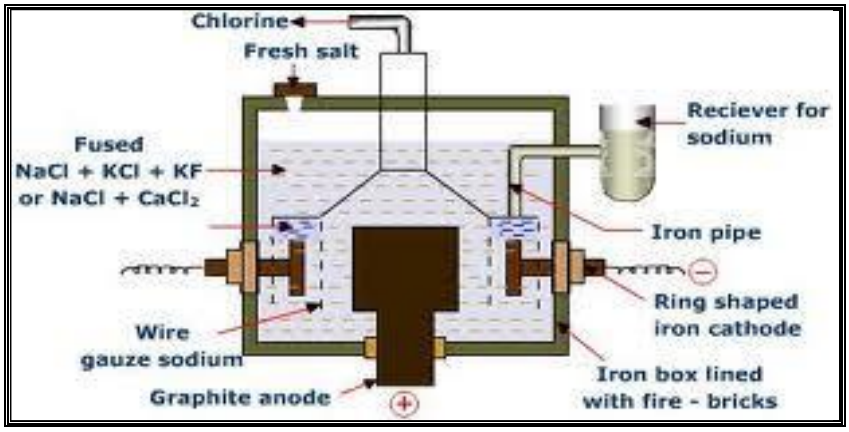
(v) Reduce visibility: Air pollution may cause smog in air which reduces visibility.

12. Required to explain the mechanism of extracting sodium from its ore.

Mechanism of extraction

Sodium is extracted by electrolysis of molten sodium chloride. The process is carried out in the Down's cell. Calcium chloride is added to the cell to lower the melting point of sodium chloride from about 774℃ to about 600℃. ***01 Mark***

Down's cell ***02 Marks***



A very high current about 30,000A is used to keep the sodium chloride in molten state, allowing Na^+ and Cl^- to move freely.

At cathode

Sodium ions gain electrons to form sodium atoms which float on the electrolyte and collected as molten sodium in the receiver. ***01 Mark***

$Na^+(aq) + e^-(aq) \longrightarrow Na(s)$ ***01 Mark***

At anode

Each chlorine ion loses an electron and combines with another chlorine atom to form chlorine gas. The gas is collected as a useful product. ***01 Mark***

$2Cl^-(aq) \longrightarrow Cl_2(g) + 2Cl^-(aq)$ ***01 Mark***

13.(a) Required to name the function groups

(i) R – COOH Carboxyl group ***01 Mark***

(ii) R – COOR Esters ***01 Mark***

(iii) R – OH Hydroxyl group ***01 Mark***

(b) Required to complete and balance the following organic reactions.

(i) $2C_2H_2$ (g) + $5O_2$ (g) $\xrightarrow{\text{Heat}}$ $2H_2O$ (l) + $4CO_2$ (g) ***01 Mark***

(ii) $CH_3CH_2OH + KOH \longrightarrow CH_3CH_2OK + H_2O$ ***01 Mark***

(iii) $CH_3COOH + CH_3CH_2OH \longrightarrow CH_3COOCH_2CH_3 + H_2O$ ***01 Mark***

(c) (i) Methane reacts with chlorine in the presence of ultraviolet light to form tetrachloro methane. The reaction takes place in four stages.

Stage 1: $CH_4(g) + Cl_2(g) \xrightarrow{\text{U.V light}} CH_3Cl(g) + HCl(g)$ ***0.5 Marks***

Stage 2: $CH_3Cl(g) + Cl_2(g) \xrightarrow{\text{U.V light}} CH_2Cl_2(g) + HCl(g)$ ***0.5 Marks***

Stage 3: $CH_2Cl_2(g) + Cl_2(g) \xrightarrow{\text{U.V light}} CHCl_3(g) + HCl(g)$ ***0.5 Marks***

Stage 4: $CHCl_3(g) + Cl_2(g) \xrightarrow{\text{U.V light}} CCl_4(g) + HCl(g)$ ***0.5 Marks***

Overall equation

$CH_4(g) + 4Cl_2(g) \xrightarrow{\text{U.V light}} CCl_4(g) + 4HCl(g)$ ***01 Mark***

(ii) Ethanol dehydrates to form ethene, then ethene hydrogenates to form ethane.

The reaction takes place in two stages.

Stage 1: Dehydration of ethanol to form ethene. ***0.5 Marks***

$$CH_3CH_2OH\ (aq) \xrightarrow[170℃]{\text{Conc } H_2SO_4} CH_2 = CH_2(g) + H_2O(l)$$ ***01 Mark***

Stage 2: Hydrogenation of ethene to form ethane. ***0.5 Marks***

$$CH_2 = CH_2(g) + H_2\ (g) \xrightarrow{\text{Nickel}} CH_3CH_3\ (g)$$ ***01 Mark***

(iii) Ethanol undergoes oxidation to form ethanoic acid. ***01 Mark***

$$CH_3CH_2OH\ (aq) + [O] \xrightarrow[H^+]{MnO_4^-} CH_3COOH\ (aq) + H_2O\ (l)$$ ***02 Marks***

14. (a) Required to define.

(i) Soil profile: is the vertical section through the soil showing soil layers from the surface to the underlying rock. ***01 Mark***

(ii) Leaching: is the process through which soluble plant nutrients are washed into the lower layers of the soil by water. ***01 Mark***

(iii) Liming: is the application of calcium in various forms to the soil in order to increase the pH of the soil. ***01 Mark***

(iv) Soil reaction: Refers to the acidity or alkalinity of the soil. ***01 Mark***

(v) Fertilizer analysis: is the percentage amount of each primary macronutrient in a fertilizer. ***01 Mark***

(b) Required any ***four factors*** that affect soil productivity apart from soil fertility. ***@ = 1 Mark***

(i) A fertile soil may not be productive due to the climatic condition of the area, which includes rainfall and temperature.

(ii) A fertile soil may not be productive due to the type of crop grown. Crops are specific to soil.

(iii) A fertile soil may not be productive due to the poor soil structure and texture.

(iv) A fertile soil may not be productive due to the poor method of farming.

(c) Required ***any six*** methods for managing the loss of soil fertility. ***@ = 01 Mark***

The loss of soil fertility can be managed through the following methods.

(i) Prevention of soil erosion: This can be done through terracing, contour farming, strip cropping and planting wind breakers.

(ii) Crop rotation: This is the practice of growing different crops on the same field at different times in orderly sequence.

(iii) Good harvesting practice: This aim to restore the fertility of the soil after harvesting. It is done by leaving plant remains to rot in the soil instead of removing them.

(iv) Controlled grazing: This involves keeping the right number of animals on a piece of land to prevent over grazing.

(v) Mixed cropping: This method involves the growing of different crops simultaneously on the same piece of land.

(vi) Use of manures: Manure comes from the decomposed organic matters. Manure binds the soil particles together to prevent soil erosion.

(vii) Agroforestry: This is the practice of planting trees together with agricultural crops to protect soil erosion after the crops are harvested.

(viii) Mulching: This is the covering of soil with a layer of organic matter such as grasses, straw, leaves and rice husks. It helps to prevent soil erosion and soil moisture.